📻 ESP32 FM Radio V2 – Final Component List

🔹 Core Electronics

Qty	Component	Specification / Notes
1	ESP32 USB-C Dev Board	30-pin version (CH340 or CP2102)
1	TEA5767 FM Radio Module	I2C interface module
1	TDA2822M Audio Amplifier IC + socket	DIP-8 package, stereo mode
1	0.96" OLED Display	SSD1306, 128x64, I2C (4-pin: VCC, GND, SDA, SCL)
2	Rotary Encoder (with push switch)	KY-040 style or equivalent
2	8Ω Speakers	0.5W minimum, 1W preferred

🔋 Power System

Qty	Component	Specification / Notes
1	18650 Battery Holder / Casing	Single-cell holder
1	3.3V Buck Converter (Mini360)	Adjustable, set to 3.3V output
OR	MCP1700-3.3 LDO Regulator	If buck converter unavailable

📦 Passive Components

Capacitors

Qty	Value	Purpose
2	470µF electrolytic	TDA2822M output capacitors

Qty	Value	Purpose
2	1µF electrolytic or film	Audio input coupling (L/R)

Resistors

Qty	Value	Purpose
2	1kΩ	Optional audio line protection

🛠 Mechanical & Build Materials

Qty	Component	Notes
1 pack	22AWG solid core wire	Power rails
Optional	Female header pins	For removable ESP32 mounting